

Published GAPSO-RF vs Our Runs

Cleveland heart disease dataset | Metric differences, feature selection and validation design | 23 September 2026

Reading this comparison: The published article reports a mean over five 70/30 holdout experiments (91 test cases per split). Our attachments report two distinct single 80/20 holdout tests (61 cases). Differences below are arithmetic percentage-point contrasts, **not** evidence of an improvement or a failed replication.

1. All comparable metrics reported by the article

Cleveland experiment	Accuracy %	Precision %	Recall %	AUC %
Published - 10-fold, all 13	83.17	83.72	78.26	91.30
Published - 10-fold, selected 7	87.80	89.76	82.61	92.00
Published - 70/30, all 13	89.01	87.97	84.78	91.00
Published - 70/30, selected 7	95.60	97.44	92.68	94.00
Our fast - 80/20, all 13	88.52	83.87	92.86	96.10
Our longer - 80/20, selected 6	83.61	78.12	89.29	92.42

Published values: El-Shafiey et al. (2022), Table 16, Cleveland dataset. Local fast values: evaluation(1).json. Local longer run: retested_holdout_metrics(1).json. AUC percentages equal 100 x ROC-AUC. The article table does not give specificity, F1, or a confusion matrix for its four Cleveland conditions.

2. Difference from the published selected-feature holdout

Metric	Published 70/30, 7 FS	Fast 80/20, 13 FS	Delta fast	Long 80/20, 6 FS	Delta long
Accuracy	95.60%	88.52%	-7.08 pp	83.61%	-11.99 pp
Precision	97.44%	83.87%	-13.57 pp	78.12%	-19.31 pp
Recall / sensitivity	92.68%	92.86%	+0.18 pp	89.29%	-3.39 pp
ROC-AUC	94.00%	96.10%	+2.10 pp	92.42%	-1.58 pp
Specificity	Not reported	84.85%	Not calculable	78.79%	Not calculable
F1	Not reported	88.14%	Not calculable	83.33%	Not calculable

Interpretation: The fast run has 2.10 percentage points higher reported AUC than the published selected-feature holdout, while its accuracy is 7.08 points lower. The longer six-feature run has 1.58 points lower reported AUC and 11.99 points lower accuracy. These directional differences may be due to splitting, thresholding, training, or model configuration. One test split cannot establish which model generalizes better.

3. What the confusion matrices independently verify

Our run	TN	FP	FN	TP	Accuracy	Specificity	F1
Fast, 13 features	28	5	2	26	54/61 = 88.52%	28/33 = 84.85%	52/59 = 88.14%
Long, 6 features	26	7	3	25	51/61 = 83.61%	26/33 = 78.79%	50/60 = 83.33%

The published paper has no corresponding confusion matrix in Table 16. AUC cannot be reconstructed from these matrices without per-patient probability scores; the local AUCs are taken from the exported JSONs. Both local matrices represent 33 absent and 28 disease-positive cases, but their row-level predictions were not supplied for a paired comparison.

4. Feature sets: exact overlap and reduction

Study / run	Selected predictors	Retained	Reduction
Published GAPSO-RF	cp, fbs, restecg, exang, slope, ca, thal	7/13	46.15%
Our fast run	age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, oldpeak, slope, ca, thal	13/13	0%
Our longer run	sex, cp, fbs, slope, ca, thal	6/13	53.85%

The published seven-feature set and our longer six-feature set share **cp, fbs, slope, ca, thal** (five predictors). The published set additionally includes restecg and exang; ours instead includes sex. Jaccard overlap = $5/8 = 62.5\%$. Our long run removes one more feature than the paper, but it does not reproduce the paper's selected subset.

5. Algorithm and training settings

Item	Published paper	Our fast run	Our longer run
Data and validation	303 Cleveland; stratified 10-fold or 70/30 holdout repeated five times	303 Cleveland; one stratified 80/20; seed 42	Reported recreated 80/20 holdout, 61 cases
Selection method	GA with discriminate mutation and PSO; 30 generations stated in timing section	GA-PSO fast verification; 5 generations, population 6	GA-PSO; 11 generation records, population 50; 800 cache misses
Chosen RF	Paper Table 9 gives a grid, not the winning hyperparameters	100 trees, depth 10, split 5, leaf 3, max_features sqrt, balanced	600 trees, depth 10, split 10, leaf 10, max_features 0.33, class_weight None
Decision threshold	Not specified in Table 16	0.50	Optimization log 0.45; retest JSON 0.50
Optimization duration	2,559.10 s in Table 17 (experiment context differs)	1.91 s	3,360.24 s (56.00 min)

Published RF candidate grid (Table 9): trees 50/100/200/500/1000/2000; depth 3/5/10/15; min split 2/5/10/20; min leaf 1/5/10/12; max_features auto/log2/sqrt. Our longer winning RF has 600 trees and max_features 0.33, which are absent from that grid. Runtime values are from different hardware and tasks and should not be presented as a speedup or slowdown.

6. Conclusions suitable for a research manuscript

Supported: The local fast RF correctly classifies 54/61 and the longer six-feature RF correctly classifies 51/61 in their respective exported holdout reports. The longer run selects six predictors versus seven reported by El-Shafiey et al. The paper reports substantially higher repeated 70/30 holdout accuracy than either local single-split result.

Not established: superiority or inferiority of any approach, a matched reproduction of the article, clinical validity, or significance of AUC differences. Our two local runs also differ in feature sets and RF settings, and the long optimization threshold 0.45 conflicts with the retest threshold 0.50.

To obtain a fair difference: rerun the paper-style 10-fold CV and repeated 70/30 protocol on the same UCI Cleveland data with training-fold-only imputation and feature selection. Save per-split metrics, selected features, test indices, model versions and test probabilities for both implementations. Separately compare models on one identical, genuinely untouched 80/20 holdout.

Sources

El-Shafiey MG, Hagag A, El-Dahshan EA, Ismail MA. A hybrid GA and PSO optimized approach for heart-disease prediction based on random forest. Multimedia Tools and Applications 81 (2022), 18155-18179. doi:10.1007/s11042-022-12425-x. Values used here: Tables 9, 15, 16, 17 and section 4.2. Source PDF: A_hybrid_GA_and_PSO_optimized_approach_for_heart-d(2).pdf.

Local exports: evaluation(1).json; metrics(1).json; retested_holdout_metrics(1).json; optimization_log(1).json; PIPELINE_RUN.md. Reported metrics were checked against supplied confusion matrices where possible. No raw per-patient predictions or published experiment code were provided for independent replication.